PRODUCT INFORMATION

EMULCUT FT-M

Water dilutable metalworking fluid

EMULCUT FT-M is an ester-based metalworking fluid, free from mineral oil and boron. The high lubricity provided by fatty esters, based on natural raw materials, provides an extremely high performance even in severe machining operations. It is suitable for a wide range of machining processes, including broaching, reaming, sawing, gun drilling etc. The high lubricity and EP performance also makes the product suitable for metal forming, drawing and bending operations, with excellent machine and component cleanliness.

EMULCUT FT-M emulsions are well suited for use on difficult aluminium materials such as hypereutectic aluminium alloys. As a coolant free from boron, it also serves for steel, cast iron and titanium machining processes. EMULCUT FT-M is even suitable for ductile, soft aluminium alloys in cutting, forming, bending operations, providing an excellent surface finish. Due to the efficient anti corrosion additives system, EMULCUT FT-M-emulsions provide long-term pH stability and optimum service life.

EMULCUT FT-M emulsions are recommended to be made-up using proprietary coolant mixing equipment such as a Jetmix. Alternatively, slowly add the concentrate to the water in a turbulent area of the system. The water should be 180 to 360 ppm $CaCO_3$ (10 – 20 °dH) total hardness and at room temperature. The emulsion's working temperature should not exceed 40 °C.

Key benefits:

- free from mineral oil and boron
- stable emulsion
- enhanced corrosion protection
- excellent lubricity
- suitable for machining difficult aluminium alloys
- suitable for forming , drawing operations

The concentrate should be stored at temperatures between 5 to 35 °C and protected from frost.
The shelf life of EMULCUT FT-M is 6 months, and the shelf life on containers must be observed.

Physical data

Concentrate	Appearance	visual	brown	
	Density/20 °C	DIN 51757	approx. 0.984	g/cm³
	Viscosity/20 °C	DIN 51562/1	approx. 213	mm²/s
	Pour point	ISO 3016	< -16	°C
Emulsion	pH level, 10 %	DIN 51369	approx. 9.3	
	Corrosion protection 5 %	DIN 51360/2	note 0	
	Foam attitude 5 %, water 15 °dH		Low foam, fast knock-down	
	Residues 5 %		Oily, re-emulsifiable non-sticky	

Factors for concentration determination

Refractometer	1.0
Acid split method	1.4
acidimetric titration up to pH 4	1.20*
acidimetric titration up to pH 7	1.52*

Concentrations recommended for use

Universal use/CNC machines:	5 - 8 %
Drilling, boring, milling, lowering:	5 - 8 %
Deep drilling (Sandvik Ejector boring head, lip borer):	10 - 12 %
Gear operations:	5 - 10 %
Reaming:	6 - 12 %
Chain broaching:	8 - 12 %
Sawing:	6 - 15 %
Metalforming:	10 - 20 %

*) Method used: Titration of a 10 ml sample with 0.1 n hydrochloric acid up to pH 4

Only valid in combination with EC Safety Data Sheet **160714-cs-224062**

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